Cumulativity and distributivity with implicatures and questions*

Moshe E. Bar-Lev
Tel Aviv University
mosheb l@tauex.tau.ac.il

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Abstract

Phrasal cumulativity (i.e., the assumption that predicates can be pluralized by applying Link's star operator) faces a problem with so-called 'intermediate' situations: If phrasal cumulativity is available, one would expect a sentence like *these four workers built a raft*, when uttered out of the blue, to be true if two of the workers built one raft and the other two workers built another raft, contrary to fact. In this paper I point out that such problems for phrasal cumulativity disappear once one takes into account both (i) the derivation of implicatures for sentences containing indefinites, and (ii) the effects of questions under discussion within a view that assumes phrasal cumulativity with domain restriction by 'covers' (Schwarzschild 1994; Bar-Lev 2025). I further argue that assuming phrasal cumulativity while having the effects of implicatures and questions in mind has some desired consequences. First, it straightforwardly resolves a conflict between the (apparent) absence of cumulativity with predicates like *build a raft* in upward entailing contexts and its presence when they are in downward entailing contexts (Schwarzschild 1994; Kratzer 2007). Second, on top of being able to explain why sentences like *these four workers built a raft* are not true in intermediate situations in out of the blue contexts, it can also explain why they are only marginally true in distributive situations (Dotlačil 2010, a.o.), as well as why some structurally parallel sentences are easily judged as true in distributive situations.

1 Introduction

The sentence in (1) is judged true in many situations. For instance, the workers could lift the stone individually (a distributive situation) or together (a collective situation), or be divided into two groups of two workers, each of which lifted the stone (an intermediate situation). More generally, the sentence seems true whenever the plurality of four workers can be divided into (potentially overlapping) parts each of which lifted the stone.

(1) The four workers lifted the stone. **Cumulativity**: ✓

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The fact that (1) is judged true in intermediate situations in particular (and does not require a special context in order for that to happen, see Schwarzschild 1994, 1996; Heim 1994) has been taken to show that the predicate *lift the stone* is 'cumulative', as in (3): If it is true of two objects, then it is also true of their sum.

(2) **Cumulativity**:
A predicate Pred is cumulative iff for any A, B: $[\![\text{Pred}]\!](A) \wedge [\![\text{Pred}]\!](B) \Rightarrow [\![\text{Pred}]\!](A \sqcup B)$

(3) A lifted the stone $\wedge$ B lifted the stone $\Rightarrow$ $A \sqcup B$ lifted the stone

There has been some debate about the source of cumulativity, and particularly whether there is a place in grammar for cumulativity operators that close the denotation of predicates under sum. Some argued that cumulativity is a property of lexical predicates (Lexical Cumulativity, henceforth LC; Krifka 1992; Landman 1996; Champollion 2016, a.o.): In the case of (1), the predicate *lift* is 'born plural', that is, it is cumulative from the get go.[1] Given that *lift* is cumulative, when it combines with the referential object *the stone* the result is also cumulative. Others argued that LC is not enough, and that Phrasal Cumulativity (henceforth PC) is needed (Sternefeld 1998; Sauerland 1998; Beck & Sauerland 2000, a.o.): Pluralization of phrasal predicates with Link's star operator (and its n-place counterparts), which closes the denotation of predicates like *lift* or *lift the stone* under sum.[2]

(1) alone does not let us distinguish between LC and PC, since *lift the stone* is expected to end up as cumulative either way. But it has been observed that in many cases that do distinguish between the two options, PC overgenerates unattested readings whereas LC makes correct predictions. Specifically, it's been pointed out that when a predicate does not combine with a referential object, PC predicts truth in intermediate situations, contrary to fact. Consider for example (4), where the predicate contains an indefinite; as observed by Link (1998); Champollion (2016), sentences like (4) do not seem true in intermediate situations, e.g., where two workers built one raft and the other two built another raft.[3]

(4) The four workers built a raft. **Cumulativity**: ✗

(5) A built a raft $\wedge$ B built a raft $\not\Rightarrow$ $A \sqcup B$ built a raft

1 I rely on the common assumption in (i) (see, e.g., Krifka 1986; Kratzer 2007), with which the notion of cumulativity with two-place predicates falls under the general characterization of cumulativity in (2), and can be stated as in (ii):

(i) $\langle a, b\rangle \sqsubseteq \langle c, d\rangle$ *iff* $a \sqsubseteq c \wedge b \sqsubseteq d$

(ii) A two-place predicate Pred is cumulative iff for any pairs $\langle A, B\rangle$, $\langle C, D\rangle$:
$[\![\text{Pred}]\!](\langle A, B\rangle) \wedge [\![\text{Pred}]\!](\langle C, D\rangle) \Rightarrow [\![\text{Pred}]\!](\langle A, B\rangle \sqcup \langle C, D\rangle)$

2 Throughout most of the paper I will talk about LC and PC as if they are mutually exclusive, since assuming PC with no restriction on its applicability makes LC redundant. Kratzer (2007) argues that one needs both PC with restriction to certain environments and LC. I will return to her arguments in section 7.2, where I will argue that the view proposed in this paper allows for assuming unrestricted PC, thus making LC redundant.

3 I use here *built a raft* rather than *lift a stone* which would be more similar to (1), since with *lift a stone* one could also consider intermediate situations that don't involve co-variation between workers and stones, i.e., where each group of two workers lifted the same stone. Situations with no co-variation would be irrelevant to our discussion, because *the workers lifted a stone* is expected to be true in these situations both if one assumes LC of *lift* and if one assumes PC, so they won't help us distinguish between the two options. See also fn. 15.

		lift the stone	*weigh 50kg*	*build a raft*
Empirical picture	Cumulative?	✓	✗	✗
Theoretical picture	Can it be cumulative due to LC?	✓	✗	✗

Table 1 Summary of how LC explains the data in (1), (4) and (6) (following Champollion 2016).

A similar problem for PC arises with predicates that involve measurement like *weigh 50kg* in (6), as observed by Lasersohn (1989). If PC were possible, one would presumably expect (6) to have a reading on which it is true whenever the boxes can be divided into groups each of which weighs 50kg. This is clearly not the case, as the fact that the sentence is not judged true in an intermediate situation demonstrates: If two of the four boxes weigh 50kg together, and the other two boxes do too, (6) looks simply false.[4,5]

(6) The four boxes weigh 50kg. **Cumulativity**: ✗

(7) A weigh 50kg $\wedge$ B weigh 50kg $\not\Rightarrow$ $A \sqcup B$ weigh 50kg

The fact that (4) and (6) are not judged true in intermediate situations poses a challenge for PC. LC, on the other hand, does not expect truth in intermediate situations in either (4) or (6), due to the fact that combining a cumulative predicate (e.g., *built*) with a non-referential argument (e.g., *a raft*) is not expected to result in a cumulative predicate in the absence of PC. A natural conclusion from this state of affairs, one that is most explicitly stated in Champollion (2016), is that LC is needed, but that there is no PC. Table 1 summarizes how LC accounts for the data we have seen so far.

Something more needs to be said though once distributive situations are considered: For instance, (6) is true in a distributive situation in which each of the four boxes weighs 50kg. This is not predicted with LC alone, and it requires proponents of LC to assume that grammar furnishes a distributivity operator (following Link 1987; Roberts 1987) that is responsible for deriving distributive readings at the phrasal level (Phrasal Distributivity; henceforth PD).

In this paper I will argue that, despite appearances, PC is advantageous over LC+PD, by focusing mostly on predicates like *build a raft*, and more generally predicates that contain indefinites (unlike (1)) and do not involve measurement (unlike (6); I will return to the case of (6) shortly). My starting point will be the observation that the argument against PC from (4) is confounded, and that PC offers a rather straightforward explanation for the apparent non-cumulativity in (4) when implicature computation is taken into account. Specifically, I will point out that, on a view that assumes PC, one should expect (4)

4 Some caution is needed here: As the discussion in Gillon (1987, 1990); Lasersohn (1989); Schwarzschild (1996) makes clear, with enough contextual support it is sometimes possible to have readings for sentences like (6) that are true in intermediate situations. Note however that even in these situations the reading is not cumulative but rather distributive, albeit to non-atomic parts. Since I'm interested here in the availability of pure cumulative readings, I focus on out-of-the-blue contexts, which only make sentences true in intermediate situations if there is genuine cumulativity. I will return to the effect of context and the availability of intermediate readings in section 5.

5 Note that, unlike (6), (4) does not look entirely false in an intermediate situation. This is presumably due to a variation between predicates with respect homogeneity effects (see Križ 2015; Bar-Lev 2025). See also fn. 20.

to give rise to the implicature that the workers did not cumulatively build more than one raft; this 'no more than one' implicature excludes intermediate situations, and would make it look like the predicate is not cumulative even if PC applied.

I will argue for this view based on two challenges that arise for LC+PD and that are resolved with PC once implicature computation is factored in. First, as Kratzer (2007) observed (elaborating on a point made by Schwarzschild 1994), while LC+PD appears to make the right cut between (1) and (4), it makes wrong predictions when they are embedded under negation: In both (8) and (9), the only available interpretation (in out of the blue contexts; see fn. 36) is one that is false regardless of whether the situation is distributive, collective, or intermediate; in other words, it looks like the predicate in both is interpreted as cumulative. LC+PD however does not explain this: In particular, it does not explain how come the predicate *build a raft* in (9) can end up cumulative, since (as established) it can only be made cumulative with PC.[6]

(8) The four workers didn't lift the stone. **Cumulativity**: ✓

(9) The four workers didn't build a raft. **Cumulativity**: ✓

In contrast, assuming PC with implicature computation straightforwardly explains this behavior, as implicatures are known to disappear under negation.

Second, a well established observation is that sentences like (4) are easily interpreted as true in collective situations, but much less so in distributive situations: If each of the four workers built a raft alone, speakers are reluctant to judge (4) as true (see Brooks & Braine 1996; Kaup et al. 2002; Frazier et al. 1999; Dotlačil 2010; Pagliarini et al. 2012; Syrett & Musolino 2013, a.o.). This is surprising if there is a freely available operator that derives readings that are true in these situations, be it with PD or with PC. However, PC with implicature computation offers a straightforward explanation for the reluctance to accept (4) as true in distributive situations: the 'no more than one' implicature that is expected to arise if one assumes PC rules out not only intermediate situations but distributive situations as well. Crucially, this implicature-based explanation for the dispereference for distributive situations is only available for PC, and is unavailable for LC+PD. These challenges are perhaps not insurmountable for LC+PD, but they are still non-trivial to resolve, whereas PC with implicature computation provides a straightforward explanation for both.[7]

This will still leave us with the problem of explaining the behavior of predicates like *weigh 50kg*, as in (6). The implicature account does not explain why, even when embedded under negation, it behaves as non-cumulative: In contrast with what we have seen for predicates like *lift the stone* and *build a raft* in (8) and (9), when *weigh 50kg* is embedded under negation, as in (10), the sentence still looks ambiguous between distributive and collective readings, neither of which rules out intermediate

6 LC+PD faces another problem here that PC does not: it has to somehow block a reading for (8) where PD applies to *lift the stone*, as this would end up true in collective situations (see Schwarzschild 1994; Bar-Lev 2025).

7 See fn. 20 for discussion of a possible account of (9) based on homogeneity effects that is compatible with LC+PD and the problems it faces, and see section 4 for discussion of some accounts of the problem with distributive situations that are compatible with LC+PD and the problems they face.

		lift the stone	*build a raft*	*weigh 50kg*
Empirical picture	Cumulative when unembedded?	✓	✗	✗
	Cumulative when embedded under negation?	✓	✓	✗
Theoretical picture	Does cumulativity persist after implicature computation?	✓	✗	(✗)
	Does a cumulative reading answer a reasonable question?	✓	✓	✗

Table 2 Summary of how PC with questions and implicatures can explain the data in (1), (4) and (6), along with their negations in (8), (9) and (10) (the approach argued for in this paper).

situations (see Heim 1994).[8]

(10) The four boxes don't weigh 50kg. **Cumulativity**: ✗

One may wonder then whether PC is still at a disadvantage compared to LC+PD when predicates like *weigh 50kg* are considered. However, in Bar-Lev (2025) I pointed out that the absence of cumulativity in (6) and (10) is not necessarily a problem for PC either, since there is an independent reason why it should not have a cumulative reading, based on the assumption that sentences should provide answers to reasonable questions under discussion (QUDs; see Roberts 1996, a.o.); when they aren't, I proposed there, distributive and collective readings can be derived by domain restriction with 'covers' (following Schwarzschild 1994; Heim 1994). The picture that emerges is one where there are two sources for non-cumulativity: One due to questions, responsible for the non-cumulativity of *weigh 50kg*, and another due to implicatures, responsible for the non-cumulativity of *build a raft* when unembedded. Table 2 summarizes this PC-based view (compare with the LC-based view in Table 1).

I will argue that taking these two sources for non-cumulativity into account helps explain not only the behavior of the various predicates we mentioned so far, but also some other intricate issues: (i) Why there is an intuitive difference between intermediate and distributive situations for sentences like (4), despite the fact that the 'no more than one' implicature for a cumulative reading of (4) excludes both of them, and (ii) why some sentences that are structurally parallel to (4), such as (11), are easily interpreted as true in distributive situations.

(11) The girls are wearing a black dress.

The paper is stuctured as follows. In section 2 I argue that PC expects implicature computation to rule out both intermediate and distributive situations for sentences like (4). In section 3 I provide evidence for this view by showing that the non-cumulativity of (4) behaves like an implicature: it is cancellable, it disappears under negation (as we have seen in (9)), and it can be targeted by 'metalinguistic' negation. In section 4 I point out that this view can further explain in a unified fashion some experimental findings concerning children's behavior and processing effects with sentences like

8 Note that, if it were for (6) alone, an implicature-based account could suffice; the fact that things do not change under negation suggests that something else is at play here.

(4). Section 5 presents the QUD-based theory from Bar-Lev (2025), and section 6 points out that combining this view with the implicature account proposed in section 2 has some desired consequences pertaining to both the difference in judgments between intermediate and distributive situations for (4) as well as the difference in judgments between (4) and (11) in distributive situations. Section 7 discusses some further issues and section 8 concludes.

2 Phrasal Cumulativity with implicatures

Consider (12), which is minimally different than (4) in that it has a singular rather than plural subject.

(12) Sue built a raft.

Notably, (12) has the implicature that Sue only built one raft. A simple explanation for this implicature is based on the assumption in (13), according to which (12) has (14) as an alternative (see Fox 2007; Spector 2007):[9]

(13) $Two\ NP_{pl} \in Alt(A\ NP_{sg})$

(14) Sue built two rafts.

In a situation where Sue built two rafts, both (12) and (14) are true, but since (14) is stronger than (12), (12) has the implicature according to which (14) is false:[10]

(15) $\mathscr{E}xh$(Sue built a raft) $\Rightarrow \neg$ Sue built two rafts.

Let us now turn to the plural example we are interested in, in (16), repeated from (4):

(16) The four workers built a raft.

Let us accept the premise of PC, namely that (4) has (or at least can have) a reading where the predicate *build a raft* is cumulative. What implicatures do we expect this sentence to give rise to? Given what we have just said for (12), (16) should have the alternative in (17), and we should expect the implicature in (18):

(17) The four workers built two rafts.

(18) $\mathscr{E}xh$(The four workers built a raft) $\Rightarrow \neg$ The four workers built two rafts.

9 Here I assume with Fox that the relevant alternative to singular indefinites is one where $A\ NP_{sg}$ is replaced with $two\ NP_{pl}$; nothing changes if one takes it to be $several\ NP_{pl}$ as in Spector (2007). One could consider making the bare plural alternative *Sue built rafts* responsible for the 'no more than one' inference, but this would only work if it's taken together with its so-called multiplicity inference, which has been argued not to come from the basic semantics of bare plurals (Sauerland et al. 2005; Spector 2007; Zweig 2009; Mayr 2015, a.o.); in fact, for Spector (2007), the multiplicity inference of *Sue built rafts* is itself derived based on the 'no more than one' implicature of (12).

10 I use here $\mathscr{E}xh$ as a place holder for whatever it is that's responsible for implicature computation, with no commitment to a particular theory of how this should be carried out.

Now consider an intermediate situation, one where the workers were divided into two groups of two workers each, and both of these groups built a raft. In this situation, (17) looks true (Scha 1984; Link 1983; Krifka 1986, a.o.), so we may expect that negating (17), as in (18), would rule it out. In other words, even if we assume PC and thus expect truth in intermediate situations given the basic meaning of (16), we should no longer have this expectation once implicature computation kicks in.

Of course, we need to say something about how (17) ends up true in the intermediate situation we consider to begin with. This can be done in various ways, but here I will assume that PC by Link's star operator obligatorily applies to every predicate unless it's singular. The LFs of (16) and (17) are then as in (19), where the semantics of the star operators is as in (20):[11, 12]

(19) a. LF for (16): [The four workers] [*[[**built] [a raft]]]
b. LF for (17): [The four workers] [*[[** built] [two rafts]]]

(20) a. $[\![{}^\star]\!](P)(x) = 1$ *iff* $\exists P' \subseteq \{x' : P(x') = 1\}[\bigsqcup P' = x]$
b. $[\![{}^{\star\star}]\!](P)(x)(y) = 1$ *iff* $\exists P' \subseteq \{\langle x', y'\rangle : P(x')(y') = 1\}[\bigsqcup P' = \langle x, y\rangle]$

Now the meaning derived by having the basic meaning of (19a) accompanied with the negation of (19b), paraphrased in (21), correctly rules out intermediate situations, since those make (19b) true.[13]

(21) $\mathscr{E}xh((19a)) = (19a) \wedge \neg(19b)$
≈ Each of the four workers participated in building a raft (alone or together with other workers), and it is not the case that the four workers built more than one raft between them.

11 Here I follow Schwarzschild (1994); Sauerland (1998); Kratzer (2007) in assuming that PC obligatorily applies quite pervasively. Specifically, I assume (with Sauerland 1998, but contra Kratzer 2007), that it obligatorily applies to all predicates (though presumably with the exception of predicates that are morphologically singular, otherwise it becomes difficult to account for the difference between singular and plural NPs). This is done in order to ensure that not only the whole VP is pluralized with * (as one would expect on Kratzer's view), but also the verb (*built*) is pluralized with ** whether the object is plural (as in (19b)) or not (as in (19a)). While it is not strictly necessary for my proposal to assume that ** applies to *built* in (19a) (in fact it ends up semantically vacuous), this makes things somewhat simpler. Particularly, if I assumed with Kratzer (2007) that only sisters of plural DPs are pluralized with PC, one would not expect PC to apply to *built* in (19a), and the LF for (16) would be expected to look like (i).

(i) [The four workers] [*[[built] [a raft]]]

In that case, the question would arise why (i) should have (19b) as an alternative, given that (i) is syntactically less complex than (19b), on accounts that restrict alternatives based on complexity (Katzir 2007; Fox & Katzir 2011). This issue is avoided by assuming that PC applies to every (non-singular) predicate. This is not the only way to avoid this issue: One could deny that alternatives are restricted by complexity (see recently Schwarz & Wagner 2024), and one could also assume LC on top of PC, which would make *built* cumulative regardless of whether PC applies or not (I thank Andreea Cristina Nicolae for pointing out to me the issue that arises here with complexity). Kratzer (2007) presented an argument against applying the star operator to every non-singular predicate (as mentioned in fn. 2), but as I will suggest in section 7.2, the data she brings up in support of restricting PC to sisters of plural DPs alone might itself be better accounted for if PC applies and implicature computation is taken into account.

12 In order to avoid clutter, my LFs do not have stars within DPs even though my assumptions require them. For the same reason, I leave quantifiers in object position without QRing them; I trust the reader to fill in the necessary QR or type shifting needed in order to resolve the type mismatch, without changing the relative scope of the scope-taking elements.

13 If one takes homogeneity into account (e.g., by assuming a trivalent definition of star operators, as in Bar-Lev 2025), the negation of (19b) will end up even stronger, entailing that no sub-plurality of the four workers built two rafts. For simplicity I will mostly set aside the issue of homogeneity in this paper.

This implicature excludes not only intermediate situations; it also excludes distributive situations where each of the four workers built a raft, since those also make (19b) true (under the assumption that numerals do not have the upper bound inference as part of their semantics).[14, 15] At the moment this may look both promising and concerning: promising because it can explain the reluctance of speakers to accept sentences like (16) as true in distributive situations (Brooks & Braine 1996; Kaup et al. 2002; Frazier et al. 1999; Dotlačil 2010; Pagliarini et al. 2012; Syrett & Musolino 2013, a.o.), and concerning because it is intuitively easier to judge (16) as true in distributive situations than in intermediate ones (as Keny Chatain, p.c., pointed out to me), which may look surprising if both are excluded by the same implicature. For now, I will set aside the difference between the two situations, and consider the exclusion of both as a desired consequence of PC with implicatures. I will get back to the difference between them in section 6, where I will argue that, when the role that questions have in determining the availability of various readings is factored in, we should expect speakers to dislike distributive situations but to dislike intermediate situations even more.

As it turns out, then, on PC it is entirely expected that (16) should not be judged simply true in intermediate situations once we take implicatures into account, and it further has the advantage that it predicts distributive situations to be excluded as well. Note that the expectation that intermediate and distributive situations should be excluded by implicature computation is unique to PC, and is incompatible with assuming LC+PD. On the latter, (16) has two readings: A collective one, true only in collective situations, and a distributive one, true only in distributive situations.[16] Neither of these readings is true in intermediate situations to begin with, so intermediate situations are ruled out as a matter of truth conditions rather than implicatures. As for distributive situations, they only make distributive readings true, and considering stronger alternatives with *two rafts* instead of *a raft* won't help ruling them out: If one has a distributive reading for (16) and a distributive reading for its alternative in (17), the overall meaning one would expect is one where each of the workers built a raft and it's not the case that each of the workers built two rafts, which is true in a distributive situation where each worker built exactly one raft.[17, 18]

14 Of course, (17) itself is not easily judged true in a distributive situation either, as one should expect given that it makes the stronger alternative *the four workers built four rafts* true.

15 The only intermediate and distributive situations that (16) is compatible with are ones that make (17) true, for the simple reason that the same raft cannot be built twice (see fn. 3); since those are excluded by the implicature, the only situation compatible with it is a collective one. When distributive and intermediate situations that don't involve co-variation are possible (e.g., a situation where all the workers individually lifted the same stone for a sentence like *the four workers lifted a stone*), the implicature would not exclude them (since they don't make *the four workers lifted two stones* true). Note that this coincides with the predictions of LC for such situations, so the point made in fn. 3 still holds, namely that considering these situations is not helpful for distinguishing between LC+PD and PC.

16 Even if they assume a distributivity operator that is relativized to covers and can derive truly intermediate readings, as in Champollion (2016), this has to be restricted to contexts where intermediate readings can arise, and thus doesn't affect 'out-of-the-blue' contexts like the ones we currently discuss.

17 Note that the overall meaning is slightly stronger with an embedded implicature in the scope of a distributivity operator, but it still ends up true in this situation.

18 Of course, much depends on the choice of alternatives here, and if one considers alternatives that aren't stronger than the sentence one could rule out distributive situations for distributive readings as an implicature, but that would face its own problems. Particularly, one may consider assuming that the distributive reading of (16) has an alternative with no distributivity operator, that is, (ia) would have (ib) as an alternative:

If the fact that (16) is not simply true in distributive and intermediate situations is due to a scalar implicature, we expect it to have the hallmarks of scalar implicatures in general. In the next section I will point out that this is indeed what we observe: (i) This implicature is cancellable, (ii) it disappears under negation (as we have seen in (9)), and (iii) it can be targetted by a 'meta-linguistic' negation. Since LC+PD is incompatible with a view where distributive and intermediate situations are ruled out by a scalar implicature, these facts provide an argument in favor of PC over LC+PD.[19]

3 Further evidence for Phrasal Cumulativity with implicatures

3.1 Cancellability

If the apparent non-cumulativity of (16) comes about due to a basic cumulative meaning derived by PC with an implicature on top, we expect it to show other known properties of implicatures. Let us first consider cancellability. The implicature from *Sue built a raft* that she didn't build two rafts can be cancelled (suspended): A speaker who is not certain whether Sue built one or two rafts can truthfully utter (22).

(22) Sue built a raft, and perhaps she even built two rafts.

We expect to see a parallel effect with a plural subject, that is, that a speaker should be able to truthfully utter (23) if they are not certain whether the workers built a raft together (a collective situation) or were divided into two groups each of which built a raft (an intermediate situation).

(23) The four workers built a raft, and perhaps they even built two rafts.

This indeed looks possible, as expected. In contrast, it is difficult to reconcile this fact with LC: Since it does not expect *build a raft* to give rise to truth in intermediate situations to begin with, it cannot

(i) a. [The four workers] [^{D}[[built] [a raft]]]
b. [The four workers] [[[built] [two rafts]]]

This would not help making the exclusion of intermediate situations a matter of implicatures (which I will argue for in the next section), but it will at least derive the exclusion of distributive situations as an implicature. This is because on LC+PD, (ib) is true when two groups of two workers each built a raft, due to *built* being lexically cumulative, and consequently negating it would rule out distributive situations (assuming that it is possible to derive contextually contradictory implicatures; see Magri 2009). However, on this possibility it is surprising that distributive readings are sometimes entirely natural for sentences similar to (16), as we will discuss in section 6 (see especially ex. (56)). As I will point out there, on the view proposed in this paper the implicature that rules out distributive (and intermediate) situations is only expected to arise for cumulative readings, and when distributive readings are possible it is not expected. But if distributive readings are themselves associated with an implicature that rules out distributive situations (even when that leads to a contextual contradiction), it becomes entirely unclear how distributive readings that allow co-variation can ever be possible for sentences similar to (16). In other words, excluding distributive situations this way would defeat the purpose of positing PD to begin with.

19 Dotlačil (2010) proposes an implicature-based explanation for the problem with distributive situations within a view that assumes PD, but his implicature is derived at the level of disambiguation and is thus of a different kind than scalar implicatures; as a consequence, the predictions we discuss in the next section are not necessarily shared with his account (most importantly, his account does not explain the **?**/Kratzer observation that cumulativity pops up with embedding under negation). See section 4 for further discussion of his account.

account for the possibility of a speaker who commits to the truth of *the four workers built a raft* without necessarily committing to non-intermediate situations.

Similarly, a speaker who isn't certain whether the workers built a raft together (a collective situation) or each built a raft individually (a distributive situation) can truthfully utter (24), which supports the view in which the source of the reluctance to judge (16) as true in distributive situations is an implicature.

(24) The four workers built a raft, and perhaps they even built four rafts.

3.2 Embedding under negation

Another known property of implicatures is their disappearance under negation: (25) is false both if Sue built exactly one raft and if she built more than one.

(25) Sue didn't build a raft.

As discussed in section 1, Kratzer (2007) pointed out that, when embedded under negation, as in (26) (repeated from (9)), predicates like *build a raft* have a cumulative meaning: (26) is judged false regardless of whether the four workers built a raft individually (a distributive situation), together (a collective situation), or were divided into two groups each of which built a raft (an intermediate situation).

(26) The four workers didn't build a raft. **Cumulativity**: ✓

This is surprising for LC+PD, which predicts that the predicate should not be able to become cumulative, but is entirely expected if PC applies under negation (as expected given the assumption that it applies to every non-singular predicate): Since non-cumulativity in (16) is due to a basic cumulative reading accompanied by an implicature, when it is embedded under negation we expect the implicature to disappear, just like in (25), and consequently only have the basic cumulative reading. Kratzer's argument for PC over LC from (26) is then maintained.[20]

3.3 'Metalinguistic' negation

While implicatures normally disappear under negation, they can be derived under negation with a particular prosody that involves stress on the implicature trigger ('metalinguistic negation'; see Horn 1985, and more recently Fox & Spector 2018; Bassi et al. 2021, a.o.), as in (27) (uppercase indicates pitch accent):

(27) Sue didn't build A raft, she built TWO!

20 One may consider an account of (26) based on homogeneity effects (Löbner 1987; Schwarzschild 1994; Križ 2015, a.o.) as a way to salvage LC+PD; specifically, one may want to explain why (26) is not judged true in distributive and intermediate situations by building on the observation that it shows homogeneity, i.e., that it isn't true even when only some of the workers built a raft. But this may not be of much help, since a theory that relies on LC+PD faces non-obvious problems in explaining homogeneity effects, specifically why some predicates appear to behave as homogeneous and some don't (see fn. 5 above), whereas this can be explained if one takes homogeneity effects to be tightly connected to PC (see Bar-Lev 2025 for discussion).

(27) can be uttered by a speaker who believes that *Sue built a raft* is true but its implicature is false. Similarly, (28) can be uttered by a speaker who believes that *the four workers built a raft* is true but its implicature is false, e.g., a speaker who believes they are in an intermediate situation where two workers built a raft and the other two did too; and (29) can be uttered by a speaker who believes they are in a distributive situation where each of the four workers built a raft:[21]

(28) The four workers didn't build A raft, they built TWO!

(29) The four workers didn't build A raft, they built FOUR!

4 Evidence from processing effects and children's behavior

In the previous version I argued for PC with implicatures based on the observation that the exclusion of distributive and intermediate situation for sentence like (16) behaves as one would expect from an implicature taht is triggered by the indefinite. In this section I will argue that this account can further explain some previous experimental findings, one concerning children's behavior and the other concerning processing effects, and moreover do so in a unified fashion, something that looks out of reach for previous accounts.

The first finding concerns children's behavior with respect to distributive and collective situations. Several experimental studies have shown that children are much more tolerant to distributive situations than adults are (see Pagliarini et al. 2012; Syrett & Musolino 2013; de Koster et al. 2018, 2021). Pagliarini et al. (2012), for example, found that while adults' rate of true judgments for sentences like (16) was at 50%, in contrast with 98% in collective situations, the rate for children up to age 9 was above 90% in both distributive and collective situations. Children are known to be different from adults in computing implicatures: At least when computing the implicature involves lexical retrieval of alternatives, they tend to compute them less than adults do (see Barner & Bachrach 2010; Barner et al. 2011; Singh et al. 2016, a.o.). On the view proposed here, where the problem with distributive situations is the result of an implicature derived by considering the alternative where *A* NP_{sg} is replaced with *two* NP_{pl}, this difference between adults and children is expected.[22]

The second finding concerns processing effects. Frazier et al. (1999) have shown experimentally that, for predicates similar to *build a raft*, reading times are longer when participants encounter post-verbal *each*, as in (30a), than when they encounter post-verbal *together*, as in (30b), and no such effect is found when they encounter *each* and *together* pre-verbally, as in (30c)–(30d).

21 (27), (28) and (29) are not entirely natural, presumably due to the fact that the indefinite article does not bear stress very easily. But note that the argument here can be replicated with more natural variants of (27) and (28):

(i) Sue didn't build ONE raft, she built TWO!

(ii) The four workers didn't build ONE raft, they built TWO/FOUR!

22 Note that one should likewise expect children to be different than adults in accepting sentences like (16) in intermediate situations; to the best of my knowledge this hasn't been tested.

(30) a. Jackson and Beverly painted a room **each** over the long weekend.
b. Jackson and Beverly painted a room **together** over the long weekend.
c. Jackson and Beverly **each** painted a room over the long weekend.
d. Jackson and Beverly **together** painted a room over the long weekend.

Frazier et al.'s interpretation of these results was that *painted a room* is by default interpreted collectively, and post-verbal *each* requires costly reanalysis—moving from a collective reading to a distributive one. In contrast, post-verbal *together* does not require reanalysis at all because it is compatible with the default collective reading. And since the choice between the distributive and collective readings is done when encountering the predicate *paint a room*, pre-verbal *each* and *together* give rise to no clash since they are encountered before the predicate is.

The view proposed here suggests an alternative explanation: *painted a room* is by default interpreted as cumulative and gives rise to the 'no more than one' implicature (see e.g. Sun & Breheny 2019 for the claim that implicatures are computed immediately in upward entailing contexts), and post-verbal *each* requires moving from a cumulative reading that excludes distributive situations (due to the implicature) to a distributive one that is only compatible with distributive situations. In contrast, post-verbal *together* requires moving from a cumulative reading that is only compatible with collective situations (due to the implicature) to a collective reading, a move that one would arguably expect to be less costly since both readings are compatible with the same situations. And pre-verbal *each* and *together* give rise to no slowdown in reading times because they are encountered before the indefinite is, hence before the implicature is computed.

This view makes the following prediction: If there is no indefinite to begin with, there should be no slowdown in reading times since no implicature is predicted to arise. This is indeed what Dotlačil & Brasoveanu (2015) found: While they replicated Frazier et al.'s findings with sentences like (31a) (when compared to sentences with *collectively* instead of *individually*), they found no such effect for examples like (31b) where there is no indefinite.

(31) a. The girls won an award **individually** during the science fair.
b. The girls won **individually** during the science fair.

Both of these experimental findings—with children's behavior and with processing effects—have been explained within LC+PD, but the existing accounts rely on stipulations and, more importantly, are mostly unable to provide a unified account for these findings. Let me briefly discuss them before we move on to the next section. Pagliarini et al. (2012) take children's behavior to support the view proposed in Dotlačil (2010) where the issue with distributive situations is due to competition at the level of disambiguation with sentences with overt *each*. However, their account crucially relies on the stipulation that sentences with no overt *each* or *together* compete with sentences with overt *each* but do not compete with sentences with overt *together*.[23] On top of this, their account cannot explain the

23 Put differently, their account faces a variant of the so-called 'symmetry problem' in implicature computation (see Breheny et al. 2018). Note that, on the complexity-based approach to the symmetry problem in Katzir (2007); Fox & Katzir (2011), neither sentences with overt *each* nor those with overt *together* are expected to be alternatives of sentences without them,

findings in Dotlačil & Brasoveanu (2015). One may consider an account of Frazier et al.'s finding where the *each* alternative is responsible for the early commitment to collective situations; but one would expect this preference to arise for both (31a) and (31b) because both have counterparts with overt *each* which should trigger this commitment to collective situations.

Dotlačil & Brasoveanu (2015) propose an account of the difference between (31a) and (31b) within LC+PD based on the assumption that PD is costly: Since (31b) is true in distributive situations because LC makes *won* cumulative, PD is not needed for truth in distributive situations and as a result there is no slowdown when *each* is encountered in (31b); in contrast, in (31a) LC is not enough for truth in distributive situations because *won an award* is not a lexical predicate and PD is needed, which is why a slowdown occurs. Note that this too relies on an unmotivated assumption, namely that PD is costly. But even if one accepts this assumption, explaining children's behavior along the same lines would require the arguably surprising assumption that applying PD is costly for adults but not for children. The account proposed here, in contrast, can explain both experimental findings with no extra stipulations and in a unified fashion.

The only previous account of the problem with distributive situations I'm aware of that can explain both of the findings discussed here is the one found in Schwarzschild (1994), according to which indefinites tend to take wide scope relative to $\star$. On this view, children may accept distributive situations because, unlike adults, they allow for the indefinite to take narrow scope; and the indefinite taking wide scope over $\star$ is what causes the early commitment to collective situations in Frazier et al.'s paradigm, a commitment that disappears when the indefinite does. Note however that this account also relies on an otherwise unmotivated stipulation about scope preferences. Furthermore, as I will point out in section 6, it does not explain why sentences parallel to (16) are sometimes readily acceptable in distributive situations; see also Dotlačil (2010: § 3.4.1) for other problems that arise for Schwarzschild's scope-based account.[24]

Let us take stock. I proposed an explanation for the fact that positive sentences with *build a raft* applying to a plural subject are not easily judged true in either intermediate or distributive situations based on PC with implicatures. I have also argued that this account correctly predicts non-cumulativity in such examples to behave like an implicature, and that it can also explain some experimental findings in a unified fashion. In the rest of the paper I will discuss several remaining issues. First, as discussed in section 1, predicates like *weigh 50kg* do not give rise to cumulative readings in either positive or negative sentences; the implicature-based account alone does not explain this. Second, the issue with distributive situations is elusive: in some cases speakers have no difficulty judging sentences with plural subjects and VPs containing indefinites as true in such situations; we should explain then why the implicature does not arise for these examples. Third, we so far treated intermediate and distributive situations for sentences like *the four workers built a raft* on a par, but, as mentioned in section 2, there is an intuitive difference between them: this sentence seems completely out in intermediate situations, but somewhat better in distributive ones. This is surprising if the reason for excluding them is the same.

since the former are arguably syntactically more complex than the latter.

24 I thank Jakub Dotlačil for a very helpful discussion of the issues discussed in this section.

In what follows I will argue that all of these issues are resolved once we adopt the view in Bar-Lev (2025), according to which cumulative readings are the default readings, and distributive and collective readings are restricted to situations in which the cumulative reading does not provide a good answer to the QUD, in which case they can be derived by domain restriction with 'covers' (following Schwarzschild 1994; Heim 1994). In section 5 I present the QUD-based account and how it resolves the first of these issues, and in section 6 I will argue that by combining the two sources for non-cumulativity—due to implicatures and due to questions—the two other issues are resolved as well.

5 Distributivity and collectivity as QUD-determined domain restriction

5.1 Avoiding irrelevant cumulative readings with distributive and collective covers

Our focus so far has been predicates like *build a raft*, where I have argued that PC makes the right predictions despite the apparent challenge from intermediate situations. However, as mentioned in section 1, another challenge for PC arises with predicates that involve measurement, like *weigh 50kg*. For these predicates PC seems to make wrong predictions, as even in negative sentences they are non-cumulative, as we discussed for (6) and (10), repeated here as (32) and (33).

(32) The four boxes weigh 50kg. **Cumulativity**: ✗

(33) The four boxes don't weigh 50kg. **Cumulativity**: ✗

In Bar-Lev (2025) I however pointed out that this challenge to PC disappears once one considers the general requirement from sentences to provide good answers to good QUDs (following Roberts 1996 and much subsequent work; see more recently Katzir 2024), since cumulative readings for these sentences would not normally be good answers to reasonable questions.[25] If *weigh 50kg* were to be cumulative, (32) would mean roughly the following:

(34) A cumulative reading of (32):
The plurality of four boxes can be divided into parts each of which weighs 50kg.
≈ Each box participated in weighing 50kg (alone or together with other boxes)

It is however difficult to imagine a situation where this (or its negation) would be a good answer to a reasonable QUD. For instance, a cumulative reading for (6) would be a good answer to the following question, but this is not a reasonable question to consider: resolving it would not tell us much about the weight of the boxes, nor would it tell us much about anything else we might be interested in (the hash in (35) and in other questions below indicates that the question is an unreasonable candidate for the QUD).[26]

25 See Haslinger (2021) who also makes a connection between QUDs and the choice between cumulativity and distributivity. A crucial difference between her account and mine is that on her account distributive readings are the default readings whereas I will assume that cumulative readings are the default ones, an assumption that I will justify in section 5.2.

26 With this alone, one would expect it to be possible to come up with contexts in which (6) can be cumulative because the question in (35) is reasonable after all. See Bar-Lev (2025) for both an attempt to consider such contexts and for the

(35) #Can the plurality of four boxes be divided into parts each of which weighs 50kg?
≈ Did each box participate in weighing 50kg (alone or together with other boxes)?

One may consider other, more reasonable questions, such as (36) and (37), but a cumulative reading of (6) is not a good answer to either of those:

(36) Does each box weigh 50kg?

(37) Do all the boxes together weigh 50kg?

Specifically, the reading in (34) would not be relevant to the questions in (36)–(37) (assuming a standard notion of relevance, following Lewis 1988), as it cuts across answers to the question (cells in the partition induced by the question): A *no* answer to (36), for example, is compatible both with worlds where (34) is true (e.g., worlds where all the boxes together weigh 50kg) and with worlds where it is false (e.g., worlds where no plurality of boxes weighs 50kg). So a cumulative reading of (32) along the lines of (34) would be either a good (relevant) answer to an unreasonable question like (35), or a bad (irrelevant) answer to reasonable questions like (36)-(37). Either way, it is not expected to be a possible reading of the sentence. As it turns out then, the argument against PC from the non-cumulativity of (32) and (33) disappears when the requirement that sentences provide good answers to good questions is taken into account, similarly to the disappearance of the argument against PC from the apparent non-cumulativity of sentences with *build a raft* when implicature computation is taken into account that I pointed out in section 2.

Of course, (32) and (33) are not simply infelicitous, as one might expect if cumulative readings were the only available readings, in light of their inability to provide good answers to reasonable questions. Instead, they have distributive and collective readings. In Bar-Lev (2025) I follow Schwarzschild (1994) and Heim (1994) in assuming that these readings are derived by domain restriction with 'covers'.[27] Concretely, let us assume that predicates combine by predicate modification with a *Cov* variable that is assigned a cover of the domain.[28] The LF for (32) would then be as in (38).[29]

(38) [The four boxes] [* [*Cov* [weigh 50kg]]]

The meaning of (38) depends on the assignment of the *Cov* variable. What can *Cov* variables denote then? A basic constraint on the denotation of *Cov* variables is that they should be covers of the plurality predicated over, namely that their closure under sum should contain this plurality, as in (39):

observation that, once homogeneity effects are taken into account, cumulative readings for *weigh 50kg* may not be possible in any context.

27 Deriving distributive and collective readings with covers is also assumed in Schwarzschild (1996), but in combination with a version of PD rather than PC as I assume here following Schwarzschild (1994); Heim (1994).

28 The implementation of covers can be done in various ways; I assume here (with Bar-Lev 2025) the implementation in Heim (1994); Beck (2001).

29 I ignore here the internal structure of the predicate *weigh 50kg* in order to simplify things. Technically, one should expect a ** operator to apply to *weigh* based on my assumptions. This alone would give rise to irrelevant meanings, but this can be avoided by having *weigh* first combine with another *Cov* variable that denotes a pair-cover (see Schwarzschild 1996).

(39) P covers x *iff* $\exists P' \subseteq P[\bigsqcup P' = x]$

To keep things simple, let us first focus on the three types of covers in (40) (there are many more cover-types one can consider; we will return to this shortly, in section 5.3): The first is a cumulative cover, where the domain is not restricted at all; this derives cumulative readings. The second is a distributive cover, where the domain is restricted to only the atomic parts of the plurality predicated over; this cover derives distributive readings. And the third is a collective cover, where the domain is restricted to the plurality predicated over alone; this cover derives collective readings.[30]

(40) a. A **cumulative** cover ('all parts are in the cover'):
P is a cumulative cover of x *iff* $\forall y \sqsubseteq x[y \in P]$

b. **Minimal** covers:
P minimally covers x *iff* P covers x and $\neg\exists P' \subset P[P'$ covers $x]$

(i) A **distributive** cover ('only atomic parts are in the cover'):
P is a distributive cover of x *iff* P minimally covers x and $\{y : y \sqsubseteq_{AT} x\} \subseteq P$

(ii) A **collective** cover ('only the plurality itself is in the cover'):
P is a collective cover of x *iff* P minimally covers x and $x \in P$

Take now the LF in (38). The *Cov* variable in (38) can in principle be assigned any of the following covers, all of which are covers of ⟦the four boxes⟧, assuming it is the plurality $a \sqcup b \sqcup c \sqcup d$:

(41) a. A cumulative cover of $a \sqcup b \sqcup c \sqcup d$: $\left\{\begin{array}{c} a \sqcup b \sqcup c \sqcup d, \\ a \sqcup b \sqcup c, a \sqcup b \sqcup d, a \sqcup c \sqcup d, b \sqcup c \sqcup d, \\ a \sqcup b, a \sqcup c, a \sqcup d, b \sqcup c, b \sqcup d, \sqcup c \sqcup d, \\ a, b, c, d \end{array}\right\}$

b. A distributive cover of $a \sqcup b \sqcup c \sqcup d$: $\{a, b, c, d\}$

c. A collective cover of $a \sqcup b \sqcup c \sqcup d$: $\{a \sqcup b \sqcup c \sqcup d\}$

If *Cov* is assigned a cumulative cover of ⟦the four boxes⟧, namely a set that contains all pluralities of boxes such as (41a), including individual boxes, the result is a cumulative reading. But as established, this reading is a good answer only for unreasonable questions like (35), so it can be discarded. If it is assigned a distributive cover, on the other hand, namely a set that contains only the individual boxes and no other part of ⟦the four boxes⟧, as in (41b), the result is a distributive reading, which is a good answer to the reasonable question in (36). And if it is assigned a collective cover, namely a cover that contains only the plurality of four boxes and no other part of ⟦the four boxes⟧, as in (41c), the result is a collective reading, which is a good answer to the reasonable question in (37).

Deriving distributive and collective readings with covers then provides an escape hatch for sentences like (32), since such readings can provide good answers to the reasonable questions in (36) and (37). This is an instance of the general ability of domain restriction to help provide good answers to reasonable

30 What I call here cumulative, distributive and collective covers are called in Bar-Lev (2025) power, minimal-atomic and minimal-singleton covers, respectively.

questions: Restricting the denotation of the *Cov* variable in (38) to only the atomic workers or only the plurality of all workers helps the sentence provides a relevant answer to a reasonable QUD, much like domain restriction in general helps a sentence like (42) provide a relevant answer to a reasonable QUD, something which in most circumstances wouldn't be possible without restricting the domain to only those students that discourse participants might care about, e.g., the students in their department.[31]

(42) Every student came to the departmental colloquium.

The predictions we get for (32) are then as follows: Since a cumulative reading doesn't satisfy the good-answer requirement, such a reading won't be available and speakers will be faced with the choice between distributive and collective readings, that is, between assuming that the QUD is (36) or (37) and choosing a distributive or collective cover accordingly. The same considerations apply in the negative example in (33), under the assumption that a sentence is relevant whenever its negation is (see Lewis 1988), so this sentence too should be ambiguous, as is borne out.[32]

5.2 Preference for cumulative readings when they can be relevant

Now that we have an account of why predicates like *weigh 50kg* have distributive and collective readings rather than cumulative ones, let us go back to the case of *build a raft*, and consider (43), repeated again from (4), and its negation in (44), repeated from (9).

(43) The four workers built a raft.

(44) The four workers didn't build a raft.

Note first that, unlike (32), (43) would provide a good answer to a reasonable question regardless of whether it has a cumulative, distributive or collective readings, as the reasonability of all the questions in (45)–(47) demonstrates.

(45) Can the plurality of four workers be divided into parts each of which built a raft?
≈ Did each worker participate in building a raft (alone or together with other workers)?

(46) Did each worker build a raft?

(47) Did all the workers together build a raft?

Put differently, cumulative readings only provide information on which atomic individuals are part

31 This view of domain restriction as a way to avoid a failure of relevance follows the spirit of Roberts (1996): "the cooperative hearer assumes that the speaker is cooperative as well (and competent) and on this basis seeks to resolve any apparent failure of cooperativity, e.g., failure to satisfy the Gricean maxims of Quality or Relevance, presupposition failure, etc., by restricting the domain". See however Feinmann (2022) for the claim that domain restriction is unable to resolve apparent failures of quality, but rather only those of relevance, which is what I assume here.

32 Homogeneity complicates the picture, since it makes the truth conditions of a sentence and its negation non-complementary, and as a result it is no longer guaranteed that whenever a sentence is relevant its negation is too. Note however that if homogeneity is a presupposition this issue disappears, since once it is accommodated, the sentence and its negation end up complementary within the updated context set. Whether homogeneity is indeed a presupposition or not is a debate that I will not go into here (for recent arguments for and against see Wehbe 2022; Guerrini & Wehbe 2024; Bassi & Bar-Lev 2025).

of pluralities that satisfy the predicate, or, in other words, which atomic individuals participated in satisfying the predicate; the relevant distinction between predicates like *weigh 50kg* and predicates like *build a raft* then essentially boils down to the difference between participation in weighing 50kg, which is normally not relevant information, and participation in raft-building, which often is. This can explain why cumulative readings for *build a raft* are easy to access, whereas cumulative readings for *weigh 50kg* are not.

Given that all three questions in (45)–(47) are reasonable ones, one might expect that (43) should be three-way ambiguous between cumulative, distributive and collective readings, since each of these questions can be given a good answer depending on whether the *Cov* variable in (48) is assigned a cumulative cover, a distributive one or a collective one. And the same three-way ambiguity should be found for its negation in (44).[33]

(48) [The four workers] [* [*Cov* [[**built] a raft]]]

However, a crucial assumption for Bar-Lev (2025), which will play an important role here as well, is that cumulative readings are the default readings, and distributive and collective readings are only available when a cumulative reading does not provide a good answer to a reasonable question; if a cumulative reading is a good answer to a reasonable question, then it is preferred:

(49) **Prefer cumulative readings when relevant**:
Cumulative readings are preferred whenever the QUD can be taken to be one for which cumulative readings are relevant.

The assumption in (49) means that, because the question in (45) is a reasonable one, the preferred reading for (43) is a cumulative one. This assumption is needed for Bar-Lev (2025) in order to explain why (44) does not have distributive and collective readings similar to those available for (33) when uttered out of the blue. It is also needed for the proposal I argued for in previous sections: If we want to maintain that the problem with distributive situations for sentences like (43) is due to implicature computation, we need cumulative readings to be preferred when they provide a good answer to the QUD, because (as mentioned in section 2) the implicature is only expected to arise for cumulative readings, but not for distributive readings; if distributive readings were freely available, it would be difficult to explain why there is any problem with distributive situations. My proposal is then committed to the assumption in (49) as well.

Let me elaborate a little on this assumption and what may justify it. Note that a cumulative cover is the only kind of cover that effectively does not involve domain restriction: Having a cumulative cover is like having no cover at all. In contrast, distributive and collective covers involve restricting the domain by not including certain pluralities in the cover. So we would expect general constraints on the availability of domain restriction to also constrain the availability of distributive and collective readings. In particular, we should ask how domain restriction behaves in general when both restricting

33 Presumably *built* in (48) also combines with a *Cov* variable (denoting a 'pair-cover') before the ** operator applies, but I ignore it here for simplicity.

the domain and not restricting it would give rise to good answers to reasonable questions, as is the case with (43), which provides a good answer both to the reasonable cumulative question in (45) (if there is no domain restriction, i.e., with a cumulative cover) and to the reasonable distributive and collective questions in (46) and (47) (if there is domain restriction to distributive or collective covers).

With this in mind, consider again the example in (42). Suppose it is reasonable for us to be interested in whether every student in the department came to the colloquium, regardless of the subfield they are interested in (because all students are officially required to attend the colloquium), but it is also reasonable for us to be interested in whether the students who are particularly interested in Semantics came (because it was a Semantics talk). In this situation, it seems very difficult to interpret *every student came to the departmental colloquium* as restricted to only those students that are interested in Semantics, in the absence of any strong reason to assume we don't care about students who don't. In general, then, it looks like we restrict the domain as little as possible in order to satisfy the requirement that a sentence be relevant to the QUD, and if there is uncertainty about what the QUD is, we opt for the one that requires less domain restriction. We should expect then that if cumulative, distributive and collective readings are all good answers to reasonable questions, we would have a preference for a cumulative one, because it involves no domain restriction whereas distributive and collective readings do. The assumption in (49) then falls out of general considerations pertaining to domain restriction, and should not be surprising if cover choice is an instance of domain restriction.

5.3 On intermediate readings and reasonable questions

In order to complete the picture, and because this will be useful for us in the next section, we should also consider another kind of cover that I didn't discuss so far: A minimal cover that is neither collective nor distributive, what we may call an intermediate cover. Such covers have been argued to be needed in certain cases (see Gillon 1990; Schwarzschild 1996), as in (50):[34, 35]

(50) The shoes cost 20 dollars.

(50) has an intermediate reading according to which each *pair* of shoes (rather than each individual shoe) costs 20 dollars. Deriving this reading requires a cover that only contains pairs of shoes rather

34 Another type of cover one could consider is one that is neither cumulative nor minimal (e.g., a cover like $\{a, b, c, a \sqcup b\}$ for the plurality $a \sqcup b \sqcup c$). I am not aware of any argument for having such covers, and one can avoid them by assuming that cover variables can only denote minimal covers, and cumulative readings are derived in the absence of a cover variable at LF. But it is also possible that such covers are difficult to argue for simply because appropriate questions, namely questions for which readings that result from such covers are relevant, are rather difficult to come by.

35 The abundance of cover-types—and hence readings—is sometimes considered a problem for the covers-based view, as the number of possible covers grows very rapidly with the number of atomic parts of the plurality predicated over: There are 109 possible covers with 3 atomic parts, 32297 with 4 atomic parts, and 2147321017 with 5 atomic parts (see Winter & Scha 2015 and entry A003465 by OEIS Foundation Inc. 2025). This could however be brought up as a problem for domain restriction in general: The number of subsets of the domain also grows fast with domain size (though admittedly significantly less so than in the particular case of covers), but speakers still somehow manage to converge on reasonable ways to restrict the domain. Note that if the domain is only minimally restricted in order to avoid irrelevance to the QUD, as I assume here, the number of possibilities that a speaker has to consider when faced with the task of cover choice (and domain restriction in general) is in most circumstances much smaller than the almost intractable number of all possible restrictions, assuming that the number of QUDs participants in conversation consider is not the set of all possible questions.

than individual shoes. The fact that this cover choice is natural here is expected given that a cumulative reading that involves no domain restriction does not provide a good answer to a reasonable question—just like with *weigh 50kg*, participation in costing 20 dollars is usually not something we care about—and questions like *does each individual shoe cost 20 dollars?* is a much less reasonable candidate for a QUD than *does each pair of shoes cost 20 dollars?*, simply because shoes are usually sold in pairs.

As has been extensively discussed in the literature, intermediate readings are not easily available in many other cases. For instance, as discussed in section 1, (32) (=*the four boxes weigh 50kg*) is not judged true in intermediate situations, despite the fact that one can come up with a cover choice that would make it true, one that only contains pairs of boxes. This is also expected if cover choice is determined by considerations of what QUDs are reasonable, as in most circumstances it is much less likely for the QUD to be *does each pair of boxes weigh 50kg?* than *does each box weighs 50kg?*, at least in the absence of any reason to think we should be particularly interested in the weight of pairs of boxes. And, as has been pointed out by Gillon (1990); Schwarzschild (1996), when being interested in the weight of pairs of boxes is contextually reasonable, intermediate readings indeed become possible.

We then end up with the following picture: When they provide a good answer to a reasonable question, cumulative readings are preferred over all non-cumulative readings because they do not involve domain restriction, as stated in (49). Among non-cumulative readings, distributive and collective readings are preferred over intermediate readings because under 'normal' circumstances—excluding for instance the case of (50)—the former are good answers to reasonable questions whereas the latter are not. So, simplifying a lot, we can characterize the default preferences over readings that we expect as in (51).

(51) **Preferences over readings**:
Cumulative readings > Distributive/Collective readings > Intermediate readings

These preferences should of course be understood as soft preferences that only apply under 'normal' circumstances, and can easily change with the context: Crucially, when cumulative readings do not provide good answers to reasonable questions they are no longer preferred over distributive/collective readings and those become readily available (as in *the four boxes weigh 50kg*), and when intermediate questions are more likely than distributive ones they become readily available (as in *the shoes cost 20 dollars*).

In the next section I will argue that combining the two sources of non-cumulativity—from questions and from implicatures—makes some correct predictions, especially when the schematic preferences in (51) are taken into account.

6 Questions and implicatures

Recall from our discussion in section 2 that the exclusion of distributive situations due to the 'no more than one' implicature is only expected on cumulative readings derived by PC, and it's not expected on PD because it derives distributive readings rather than cumulative ones. This point applies just as well within my account: We should expect the 'no more than one' implicature derived for distributive readings, unlike those derived for cumulative ones, to not rule out distributive situations. To see why, consider the LF in (52) (repeated from (48)) and its alternative in (53):

(52) [The four workers] [* [*Cov* [[**built] a raft]]]

(53) [The four workers] [* [*Cov* [[**built] two rafts]]]

(53) entails (52) regardless of the assignment of the *Cov* variable (under the assumption that the assignment of the *Cov* variable does not vary between the sentence and its alternative), and hence the result of implicature computation is expected to be as in (54):

(54) $\mathscr{E}xh((52)) = (52) \wedge \neg(53)$

What (54) means depends on the cover assigned to the cover variable:

(55) a. If $[\![Cov]\!]$ is a cumulative cover of $[\![$the four workers$]\!]$:
$\mathscr{E}xh((52)) \approx$ Each of the four workers participated in building a raft (alone or together with other workers), and it is not the case that the four workers built more than one raft between them. [=(21)]
b. If $[\![Cov]\!]$ is a distributive cover of $[\![$the four workers$]\!]$:
$\mathscr{E}xh((52)) \approx$ The workers **each** built a raft and it's not the case that the workers **each** built two rafts.

Note that a distributive situation in which the four workers each built exactly one raft is incompatible with the overall meaning in (55a), but is compatible with the one in (55b). We should expect then that, in situations that justify domain restriction with distributive covers, that is, when distributive and collective readings become available, there should be no problem with distributive situations, because distributive situations are no longer excluded.

In other words, we expect that there should be a contrast between *the four workers built a raft* and *the four boxes weigh 50kg* in distributive situations: Since the former has a preferred cumulative reading, it should have the implicature that no more than one raft was cumulatively built by the workers, which excludes distributive situations; but because the latter does not have a preferred cumulative reading (due to the unreasonability of a cumulative question), there should be no such effect, and one should expect a simply true judgment based on the available distributive reading. I believe this prediction is on the right track, but this effect might be subtle because there might be reasons to prefer a collective reading over a distributive one (perhaps due to the collective question being more likely in certain contexts), in which case the sentence might still not be judged simply true.

This effect should be more easily detected when neither a cumulative reading nor a collective one are good answers to reasonable questions. Consider for example (56), repeated from (11), repeated from (11), an example that has often been brought up as one that is easy to judge as true in distributive situations, and in fact one that looks synonymous with the overtly distributive sentence in (57) (see

Winter 2001; Champollion 2016, 2017):

(56) The girls are wearing a black dress.

(57) The girls are each wearing a black dress.

Note that here the only QUD that makes sense among the cumulative, distributive and collective ones in (58)–(60) is the distributive one in (59): Both (58) and (60) are strange questions in out of the blue contexts.

(58) #Can the plurality of girls be divided into parts each of which wears a black dress?
≈ Did each girl participate in wearing a black dress (alone or together with other girls)?

(59) Does each girl wear a black dress?

(60) #Do all the girls together wear a black dress?

We then predict that (56) should have an easily accessible distributive reading, as attested. On other views of the issue with distributive situations, instead, it is surprising that (56) is easily judged as true in distributive situations, unlike (43). For instance, on the view following Schwarzschild (1994) where the issue with distributive situations is due to a tendency of the indefinite to take wide scope over the phrasal (distributivity/cumulativity) operator, it is not clear why this preference does not make (56) infelicitous like (61) is:

(61) #There is a black dress that the girls wear.

We further make the prediction that the problem with distributive situations should disappear for predicates like *build a raft* as well once we manipulate the context in such a way that cumulative and collective questions are not good candidates to be the QUD. Consider the following context:

(62) *Context: The kids participated in a raft-building contest, where one only gets a prize if they build a raft alone.*

a. The four kids built a raft during the contest, so each of them got a prize.

Unlike in out of the blue contexts, in (62) it looks rather easy to judge *the four kids built a raft* as true in distributive situations (as the felicity of the continuation suggests). This is expected because neither collective nor cumulative questions are good candidates for the QUD in this context, leaving only distributive questions and hence a distributive reading.[36]

Supplementing our implicature-based proposal with the questions-based perspective has another

36 As pointed out in Bar-Lev (2025), contextual manipulations of this sort are also expected to make distributive readings available under negation. This is indeed borne out:

(i) The four kids didn't build a raft during the contest, so none of them got a prize. (They did build a raft together, but that doesn't matter.)

	Cumulative reading	Distributive reading	Collective reading	Intermediate reading
Collective situation	✓	✗	✓	✗
Intermediate situation	✗	✗	✗	✓
Distributive situation	✗	✓	✗	✗

Table 3 Various readings and their compatibility with various situations once 'no more than one' implicatures are derived for a sentence like *the four workers built a raft*.

welcome result, having to do with the difference between distributive and intermediate situations. Recall from section 2 that the 'no more than one' implicature derived for cumulative readings excludes distributive and intermediate situations alike, whereas empirically there is a contrast between these two situations: *The four workers built a raft* looks entirely infelicitous in an intermediate situation, but somewhat better in distributive ones. This looks surprising if these two situations are excluded by the same implicature. With the aid of the questions-based perspective we can make sense of this. Specifically, recall the schematic preference hierarchy from (51), repeated here:

(63) **Preferences over readings**:
Cumulative readings > Distributive/Collective readings > Intermediate readings

This hierarchy suggests a way to think of the difference in judgments between collective, distributive and intermediate situations that is still compatible with assuming that cumulative readings are associated with an implicature that excludes both intermediate and distributive situations. The gist of the idea is as follows: While both distributive and intermediate situations are incompatible with the preferred cumulative reading (due to the 'no more than one' implicature), there is still a difference between them: Distributive situations are compatible with distributive readings, which are indeed dispreferred, but intermediate situations are only compatible with the even further dispreferred intermediate readings (see Table 3 for a summary of the various readings and situations they are compatible with once implicature computation is considered). The difference between the two situations can then be a byproduct of the general difference in accessibility between distributive and intermediate readings discussed in the previous section.

Concretely, there is a strategy that speakers can employ in order to derive readings that are compatible with distributive situations, which is unavailable for intermediate situations: In distributive situations, speakers may entertain the possibility that a cumulative question is not a reasonable one, which would justify opting for a distributive reading, one that provides a good answer to a reasonable distributive question; using this strategy, speakers may override the preference in (49) and end up considering a distributive reading (albeit as a dispreferred reading in the absence of any obvious reasons to avoid the default cumulative reading). This strategy is not available for intermediate situations, given the general point made in the previous section about the unavailability (under 'normal' circumstances) of questions to which intermediate readings are good answers.

7 Some further issues

7.1 Predicates with other quantifiers

My main focus in this paper has been on the interpretation of sentences with a plural in subject position and an indefinite in object position. Something more needs to be said about sentences with VPs that contain other types of quantifiers.

Let us consider first existential quantifiers that do not give rise to 'no more than one' implicatures in general, such as *at least one* and *more than one*, as in (64):[37]

(64) Sue built {at least/more than} one raft.

Since (64) does not give rise to a 'no more than one' implicature (whatever might be the reason for this; see Fox & Hackl 2006; Buccola & Haida 2020), my account does not predict its plural variant in (65) to exclude distributive and intermediate situations.

(65) The four workers built {at least/more than} one raft.

Whether or not this prediction is borne out is not entirely clear to me at this point. Note however that, even if (65) turns out to resist intermediate and distributive situations just like its counterpart with *a raft* in (43) does, this cannot be taken as argument for LC+PD over the account I proposed here, since LC+PD also predicts it to be true in these situations due to *built* being (lexically) cumulative.

Another issue arises with non-monotonic quantifiers, as in (66).

(66) The four workers built exactly one raft.

Just like its counterpart with a simple indefinite in object position in (43), (66) has the inference that the four workers did not cumulatively build more than one raft. Differently from the basic example, though, this inference is not cancelable:

(67) #The four workers built exactly one raft, and perhaps they even built two.

This may look unsurprising once one considers what happens in parallel sentences with a singular subject:

(68) #Sue built exactly one raft, and perhaps she even built two.

Things are however more complicated than that: Unlike in the singular case, one may expect that in the plural case there will be scope interaction between the non-monotonic quantifier *exactly one raft* and the star operator. If *exactly one raft* takes narrow scope relative to *, one no longer necessarily expects the 'no more than one' inference (which is crucially computed above *) to be derived by the non-monotonic quantifier; it may be derived as an implicature by negating the *exactly two* alternative,

37 I thank Danny Fox for pointing out the potential problem with these quantifiers.

but then one would expect it to be cancelable.

This may then seem to force us to assume that *exactly one raft* takes wide scope relative to ⋆ (an assumption that will face some problems; see the problems Dotlačil 2010: § 3.4.1 brings up for Schwarzschild's stipulation about scope that we discussed earlier). However, it is not in fact necessary to make this assumption. It has been independently argued that *exactly n* should be thought of as equivalent to *n*, and that what *exactly* does is force an upper bound inference to be derived higher up in the structure (see Landman 1998; Spector 2014). This will explain the behavior of *exactly one* here without having to assume that the whole DP takes wide scope, as long as the upper bound inference is forced to be derived at a position high enough to be above the star operator.

7.2 PC with predicates of times

Closely related arguments to the ones against PC with predicates of individuals have been brought up as arguments against PC with predicates of times (or events; I will not distinguish here between the two options). I will not aim to develop a full account of PC with times here, but I would still like to briefly point out that these arguments suffer from a confound similar to those that arguments against PC with predicates of individuals have, and that an account that assumes PC with implicatures on top might fare better for predicates of times as well.

Kratzer (2007) considers sentences like (69) as a problem for a free application of PC to predicates of times (see also Champollion 2016):

(69) Sue petted a rabbit for two hours.

Without going into too much detail, the argument is based on the observation (following Zucchi & White 2001; van Geenhoven 2004) that (69) is not judged true if two different rabbits are alternately petted. This is unexpected if PC can apply to a predicate that is true of times where Sue petted a rabbit, because otherwise the result would be a plural predicate of times that is true of times composed of sub-intervals each of which is an interval where Sue petted a rabbit, and the overall meaning of the sentence would be roughly as follows:

(70) There is a two-hour interval composed of intervals each of which is an interval in which Sue petted a rabbit.

In that case, one would expect it to be possible to have different sub-intervals of the two-hour interval with different rabbits petted in each, contrary to fact. Due to this problem, Kratzer assumes that PC must be restricted so that it doesn't apply to such a predicate in (69) (see Kratzer 2007 for elaboration, and see Champollion 2016 for a similar view, though one that assumes that PC is not available at all).[38]

I would like to point out that given the view proposed in this paper there is no need to restrict PC so that it doesn't apply to such predicates of times (and consequently there is no need to assume LC

38 One could consider other solutions to this problem, such as letting the indefinite take wide scope over time pluralization, or assuming a referential treatment of indefinites (see Zucchi & White 2001), but this raises questions about explanatoriness that one may wish to avoid.

on top of PC as Kratzer does). As the reader may expect at this point, this is because we should expect (69) to have the alternative in (71), which can be true in a situation where Sue petted two rabbits alternately for two hours (I will not go here into the details of how that should be derived):

(71) Sue petted two rabbits for two hours.

Implicature computation for (69) would then exclude (71), and would rule out situations in which two rabbits were alternately petted. Applying PC to the relevant predicate of times is then not harmful once implicature computation is taken into account. Furthermore, the arguments for applying PC and deriving an implicature on top of it from section 3 can be replicated here: this implicature is cancellable, as in (72), it disappears under negation—(73) is not true if two rabbits were alternately petted—and it can be embedded under negation with a 'metalinguistic negation' prosody, as in (74).[39]

(72) Sue petted a rabbit for two hours, and perhaps she even petted two rabbits for two hours.

(73) Sue didn't pet a rabbit for two hours.

(74) Sue didn't pet A rabbit for two hours, she petted TWO rabbits for two hours!

8 Conclusion

This paper argued that assuming PC makes the right predictions for sentences with plural subjects and indefinite objects once one considers the implicatures they are expected to give rise to, as well as the requirement that an utterance should provide a good answer to the QUD. I presented several arguemnts for this view. First, it can explain why sentences with predicates like *build a raft* are normally not judged as true in either intermediate or distributive situations (while also explaining the differences between them). Second, it can explain why distributive and intermediate situations are ruled out when such predicates are embedded under negation, and is more generally supported by the fact that the exclusion of these situations (in positive sentences) has the hallmarks of implicatures. Third, it can explain some experimental findings that are otherwise difficult to explain in a unified fashion concerning both processing effects and children's behavior. Fourth, it can explain why sentences containing indefinites are sometimes easily judged as true in distributive situations.

Before concluding, I would like to mention that the account developed here is incomplete in at least two respects. First, it does not attempt to provide a complete theory of the choice between cumulative, collective, distributive, and intermediate readings. I assumed here that the question under discussion affects the choice between these readings and that in the absence of good reasons to assume otherwise

39 As Champollion (2016) observes, under certain circumstances it is possible to make sentences similar to (69) true in situations that involve different objects in different sub-intervals. He considers the example in (i), and points out that it is "acceptable in a hospital context where the patient's daily intake is salient. It does not require any pill to be taken more than once."

(i) The patient took two pills for a month and then went back to one pill.

The account I sketched here is compatible with his view, where this is the result of making a particular cover salient, one that is restricted to one-day intervals.

cumulative readings are preferred. But this does not rule out the possibility that some other factors also play a role in this choice, be they pragmatic, syntactic, or some combination of the two. Developing a more comprehensive account of the interaction between syntactic structure and contextual factors in choosing between readings is something I would have to leave for future work (for some relevant experimental work on the relationship between the various readings see Poortman 2017; Glass 2018; Maldonado et al. 2017, 2019).

Second, I assumed here that cumulativity is derived by applying star operators to predicates. This has the effect of restricting cumulative readings to cases where the relevant predicates can be syntactically derived. Beck & Sauerland (2000) take this to be a good consequence of assuming PC with stars, but Schmitt (2019) argues that cumulative readings arise even when this cannot possibly be due to stars applying to syntactically derived predicates. Another problem that arises for the view I assumed here has to do with cumulative readings with singular quantifiers (see Schein 1993; Kratzer 2003; Haslinger & Schmitt 2018; Chatain 2021, 2022). Both of these issues have been argued to require more sophisticated machinery than the system I assumed here. Whether a unified account of cumulativity that is compatible with the account proposed here can be developed remains to be seen.

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